

MUSKEGON COUNTY AP, MI, USAWMO: **726360**Lat: **43.171N**Lon: **86.237W**Elev: **191**StdP: **99.06**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **14840****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-12.8	-19.2	0.7	-14.0	-17.1	0.9	-11.8	12.4	-1.2	11.4	-0.7	4.2	130	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	30.2	22.3	28.9	21.6	27.7	20.9	24.0	28.2	23.2	26.9	22.4	26.0	5.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.8	26.5	21.9	16.9	25.6	21.0	16.0	24.8	73.1	28.3	69.6	27.0	66.5	25.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.8	8.7	-19.0	32.6	3.5	1.6	-21.5	33.8	-23.6	34.7	-25.6	35.6	-28.1	36.7		
(5)			-19.4	25.4	3.4	1.3	-21.8	26.3	-23.8	27.0	-25.7	27.8	-28.2	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-3.6	-2.9	1.5	7.9	13.9	19.3	21.7	21.1	17.2	10.8	4.6	-0.5
	DBStd	10.05	5.29	5.28	5.73	4.82	4.62	3.65	3.12	2.86	4.03	4.46	4.51	4.67
	HDD10.0	1704	421	360	271	97	14	0	0	0	1	44	171	326
	HDD18.3	3608	679	593	521	316	152	31	7	9	68	238	412	583
	CDD10.0	1457	0	1	9	32	136	278	362	343	218	68	9	1
	CDD18.3	319	0	0	1	1	15	59	111	93	35	3	0	0
	CDH23.3	2332	0	0	3	8	120	459	869	624	232	16	0	0
	CDH26.7	497	0	0	0	0	26	108	211	109	43	1	0	0
Wind	WSAvg	4.2	4.9	4.7	4.5	4.7	4.1	3.6	3.4	3.3	3.5	4.3	4.7	4.8
Precipitation	PrecAvg	847	60	45	61	80	74	67	62	83	89	81	78	67
	PrecMax	1112	139	118	167	189	244	179	153	251	344	235	172	136
	PrecMin	587	9	4	9	11	8	12	22	3	4	10	13	20
	PrecStd	129	27	26	33	33	43	35	27	50	59	45	40	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	13.5	21.8	24.5	29.5	31.4	32.4	30.9	30.2	25.5	18.4	14.2
		MCWB	10.0	9.4	15.5	15.7	20.2	22.8	23.9	23.1	21.1	19.5	13.6	12.5
	2%	DB	7.4	9.7	16.6	21.4	26.4	29.4	30.4	29.1	27.8	22.8	15.9	10.1
		MCWB	5.8	7.1	12.1	13.7	18.8	21.3	22.8	22.3	20.7	17.7	12.4	8.6
	5%	DB	5.2	6.1	13.1	18.7	24.2	27.7	28.9	27.8	26.0	20.6	13.8	8.0
		MCWB	3.6	3.8	9.1	11.9	17.3	20.3	21.9	21.5	19.9	16.4	10.9	6.0
	10%	DB	3.4	4.0	9.9	16.1	22.0	26.1	27.6	26.6	24.2	18.3	11.7	5.9
		MCWB	2.2	2.3	6.6	10.4	15.8	19.5	21.2	20.8	19.0	14.8	9.1	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	11.1	16.2	18.0	21.8	24.1	25.5	25.0	23.4	20.7	15.0	12.5
		MCDB	11.4	11.7	21.1	22.6	27.7	29.8	30.3	28.9	27.5	24.1	17.0	13.8
	2%	WB	6.3	6.9	12.9	15.1	20.0	22.7	24.3	23.8	22.1	19.0	12.9	9.0
		MCDB	7.5	9.7	15.9	19.0	24.6	27.8	28.7	27.1	25.5	21.6	15.1	10.1
	5%	WB	3.8	4.0	9.7	12.9	18.5	21.6	23.4	23.0	21.0	16.9	11.1	6.4
		MCDB	4.9	5.8	12.5	17.4	22.7	26.0	27.3	26.1	24.4	19.9	13.2	7.8
	10%	WB	2.3	2.4	6.7	10.8	16.9	20.5	22.5	22.1	20.0	15.1	9.4	4.1
		MCDB	3.4	3.9	9.7	15.1	21.0	24.5	26.1	25.3	23.4	17.9	11.7	5.9
Mean Daily Temperature Range		MDBR	5.9	6.8	8.5	10.0	10.4	10.1	9.7	9.4	10.1	8.7	6.8	5.4
	5% DB	MCDBR	7.9	9.2	12.0	13.7	12.4	11.9	10.4	10.1	11.3	11.5	9.4	7.9
		MCWBR	7.0	7.2	8.3	8.1	6.9	5.8	5.1	5.0	5.7	7.3	7.2	7.2
		MCDBR	7.4	8.6	11.0	11.9	10.9	9.9	8.8	8.0	9.0	10.0	8.4	7.6
	5% WB	MCWBR	7.0	7.2	8.5	8.3	7.1	5.6	4.8	4.6	5.4	7.4	7.2	7.3
Clear-Sky Solar Irradiance	taub	0.299	0.328	0.342	0.372	0.403	0.418	0.418	0.414	0.390	0.353	0.319	0.298	
	taud	2.407	2.308	2.368	2.314	2.294	2.304	2.306	2.321	2.356	2.445	2.482	2.453	
	Ebn at Noon	835	861	898	893	871	856	851	842	836	827	807	805	
	Edh at Noon	77	100	107	123	129	128	126	121	108	87	72	68	
All-Sky Solar Radiation	RadAvg	1.20	2.03	3.56	4.75	5.58	6.25	6.35	5.41	4.30	2.65	1.53	0.97	
	RadStd	0.14	0.25	0.35	0.44	0.51	0.40	0.30	0.32	0.37	0.32	0.23	0.14	